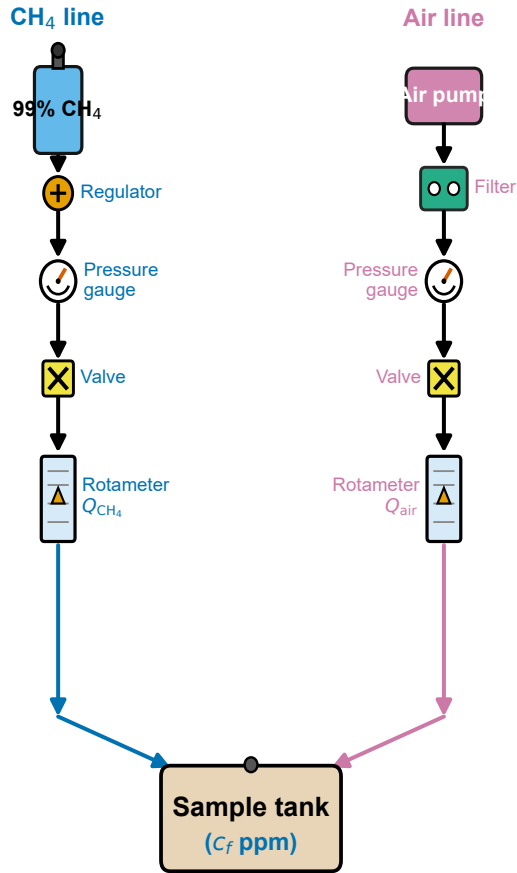


Gas mixing setup for target CH₄ concentration in a sample tank

A Dual-line mixing into sample tank



Flow tags: Q_{CH_4} (left rotameter), Q_{air} (right rotameter)

B Target concentration from two flows

Ideal-gas mixing at same T, P

Mother: $c_m = 0.99 \times 10^6 = 990,000 \text{ ppm}$ (99% CH₄)

Ambient air background: $c_a \approx 2 \text{ ppm}$ (optional)

General balance (continuous fill / same fill time):

$$c_f = \frac{Q_{CH_4} c_m + Q_{air} c_a}{Q_{CH_4} + Q_{air}} \quad [\text{ppm}]$$

If $c_a \approx 0$ (ignore ambient CH₄):

$$c_f \approx \frac{Q_{CH_4}}{Q_{CH_4} + Q_{air}} \times 990,000$$

Worked example (ignore c_a):

$Q_{CH_4} = 0.05 \text{ mL/min}$, $Q_{air} = 99.95 \text{ mL/min}$
 $\Rightarrow c_f \approx 495 \text{ ppm}$

Safety: 99% CH₄ is flammable (LEL ~5%). Use ventilated lab, no ignition sources; prefer certified dilute mother gas when possible.